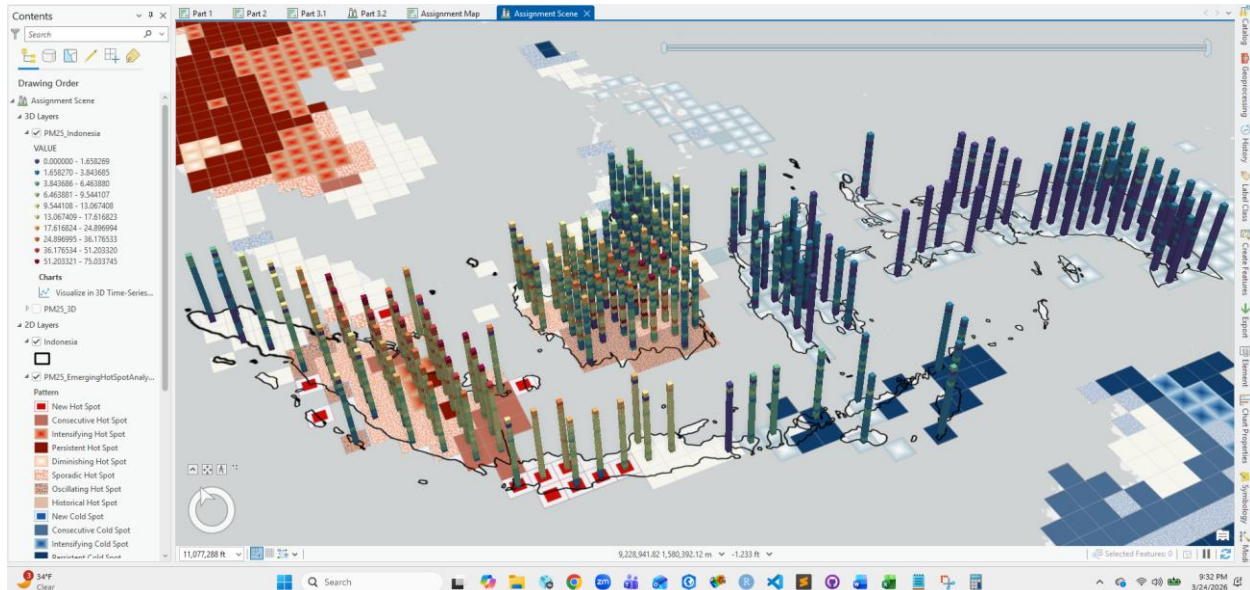


Part 1



For part one, I chose to focus on Indonesia because I saw large amounts of cold and hot spots in the emerging hot spot analysis layer. From 1998 to 2016, this data shows that consecutive, sporadic, and diminishing cold spots are prevalent on the eastern side of Indonesia. On the western side of Indonesia new, consecutive, sporadic, oscillating, persistent, and intensifying hot spots are prevalent. It also appears to be a pattern that the highest levels of PM2.5 are recorded within the last 2-3 years (time slices) of the dataset, and that (with the exception of a few areas on the southernmost region) nearly the entire country is increasing their PM2.5 levels.

Part 2

Q1) By right clicking on the fields “day_of_wk” and “hour_of_da” and then selecting visualize statistics, I see that Wednesday has the highest number of these violations, followed by Thursday and Friday. For hours, I see that the 11th hour has the highest number of these violations, followed by 12 and 13.

Q2) The default distance interval of this cube is 139 meters; this means that fish net grid boxes are 139 meters wide and long. The default time step interval is 6 months; this means that each time slice in the cube represents 6 months in progression through this dataset (starting from 03/06/2011 and ending 09/19/2019) where point values within these time and distance intervals are aggregated into bins. There are 18 time steps in total.

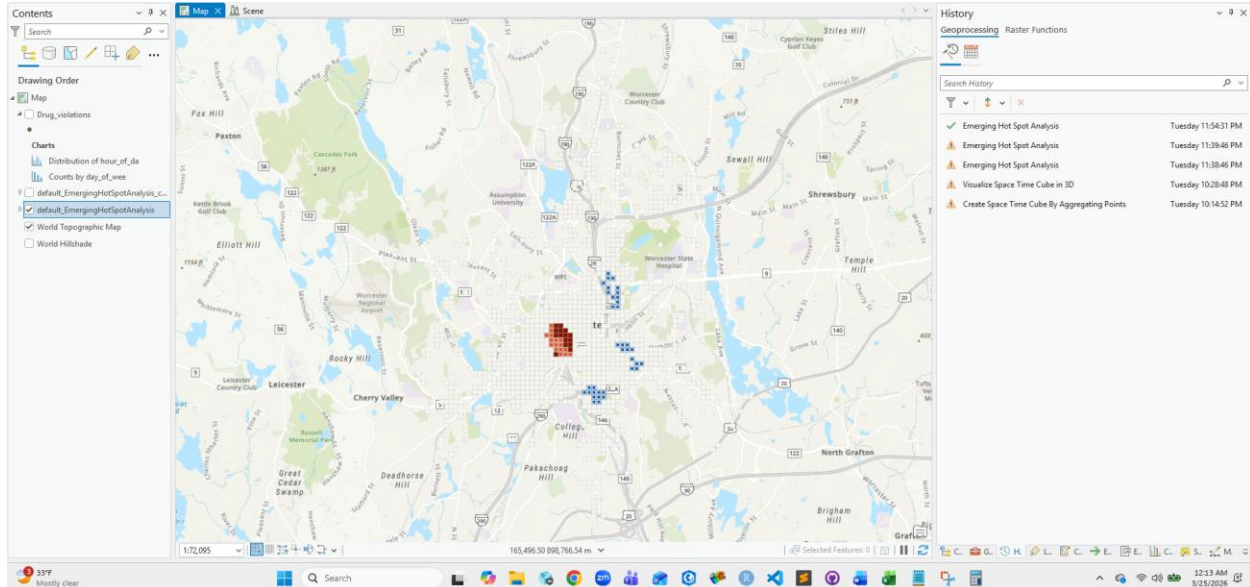
Q3) Over time, it appears that most of the city has a small amount of high-count bins scattered across space and time with trending in a clear way. However, toward the center of town and especially on Main

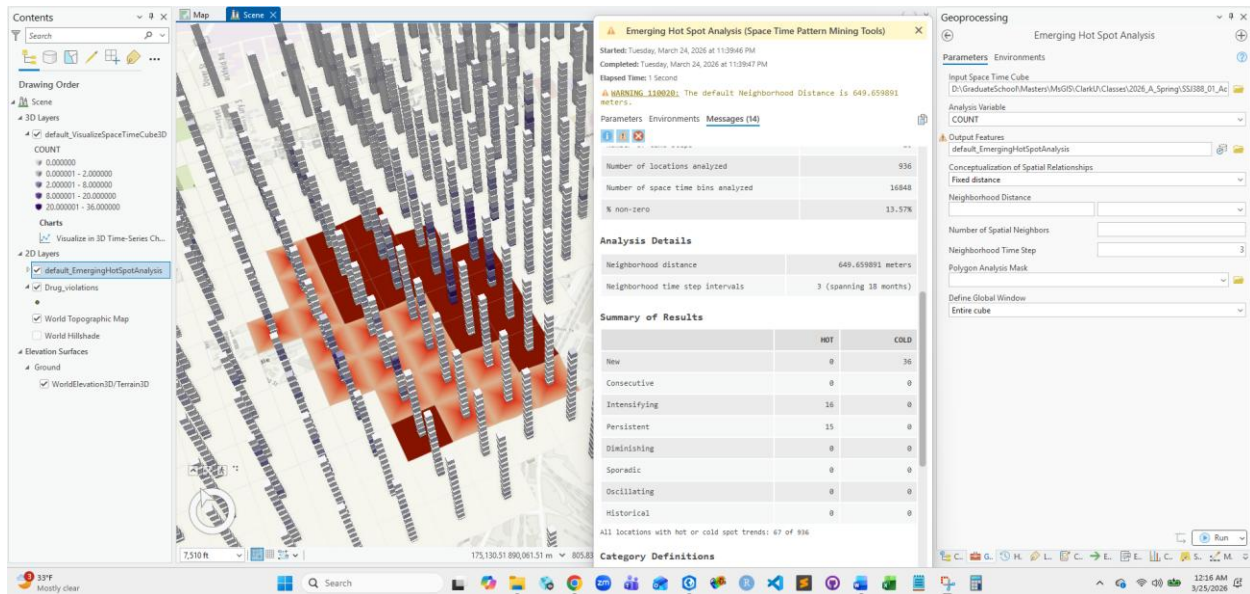
Street (south of town hall) there appears to be higher concentrations of high-count bins. Bins with counts of greater than or equal to 10 (31 out of 16,848 bins) are all in the same neighborhood around Main Street (about 1,600 meters up the street from Clark University, according to the measuring tool). These 31 bins occurred between 3/19/2011 and 3/19/2018. The time intervals with the highest number of bins among these are 3/19/2015 to 3/19/2016 with a total of 7, and there is a count total of 105 for these 7 bins.

Q4) The default neighborhood distance is 649.659891 meters. 67 out of 936 locations were identified as having hot or cold trends. There were 36 New cold spots, 16 Intensifying hot spots, and 15 Persistent hot spots. There is one cluster of hot spots which is located around Main St - between Larch St and Gold St (north/south) and Chandler St and King St (east/west). The cold spots are all located east of 290 interstate, one clustered around Venerini Academy; the second cold cluster further south is along route 122 on either side of the Hale St intersection; the third cold cluster furthest south is between Vernon Hill Park and the bend where route 146 veers south east off of 290.

Q5) There were 20 locations out of 936 identified with hot spot trends this time. With these parameters, there are no cold spots in the new analysis. Hot spot types include 10 consecutive, 4 intensifying, 3 persistent, 1 diminishing, and 2 sporadic. There is consistency in hot spot locations clustered around Main St between Federal Square and the intersection of Hammond and Main. I think this consistency is due to the high frequency of incidents among neighboring bins in this area, and there is greater contiguity of high-count bins in this area as well. Whether defining the spatial relationship through fixed distance or contiguity edges corners, this area has the highest counts.

Q6)





Above are two screenshots of the same emerging hot spot analysis result (with default conceptualization of spatial relationships) in a map and local scene. The fishnet grid aggregation shape type space-time cube was created using the “Drug_Violation” point layer provided by crimereports.com consisting of 3,650 drug violation incidents between 03/06/2011 and 9/09/2019 throughout Worcester, MA. The space-time cube is based on 139 square meter bins with a time step interval of 6 months, which means each grid location has 18 time slices. This space-time cube is then visualized based on the count of incidents that fall within each bin. The emerging hot spot analysis is based on the space-time cube bin counts with the default fixed distance (649.7 meters) set as the conceptualization of spatial relationship and neighborhood time step set at 3. This means that bins which are within 3 time intervals (18 month range) and within about 650 meters of each other are analyzed to determine hot (or cold) spot trend locations. With the above listed parameters, this analysis indicates that there is one major cluster of hot spots around the South Main neighborhood, with a few cold spot clusters on the east side of I-290.

Part 3

I chose the article titled “Spatio-Temporal Sentiment Hotspot Detection Using Geotagged Photos” because this I think this premise is an emerging concept in geospatial analysis. During a fall NEARC conference in Burlington, VT last year or the year before, the keynote speaker was a mathematician from UVM who was doing a spatial analysis on people’s use of “emoji” characters on social media (Facebook, Instagram, etc.) in correlation with current events. This article discusses using an AI/ML approach to classify geotagged and time-stamped photos based on a deep-learning trained package to interpret emotional sentiment connected to a place. I wondered how this process was able to work, and whether their results had any meaningful impact or insight for municipal or state planners to improve quality of residents/visitors experience in a certain area. Additionally, the monitoring aspect of this research is concerning to me and I wondered how this approach dealt with that.

In this analysis, the authors use two space-time tools/methods: the Getis-Ord Gi for spatial hotspot detection (also used by the emerging hot spot analysis tool in ArcGIS) and the Mann-Kendall test for spatial-temporal hotspot detection (space-time cube). Once the geotagged photos are classified based on their emotion, they want to map these photos to understand spatial distribution of public sentiment. The grid created is 1000 by 1000 with each bin measuring 12 by 14 meters – this becomes the basis of their analysis instead of the geocoded points from photographs. Getis-Ord Gi statistic is used to identify cluster trends of high and low emotion ratios spatially. Because their photos have time-stamps, trends in these ratios are able to be analyzed through time (in their case intervalled by year). First the global detection is performed using the emerging hot spot analysis tool in ArcGIS, followed by the Mann-Kendall test to identify temporal trends for each pin area (correlation between a location's Gi value and the year the photo was captured). Last, local temporal analysis is performed using the emerging hot spot analysis tool again for a select few locations where known temporal events have taken place in order to train the model on changes in emotional ratios over time.

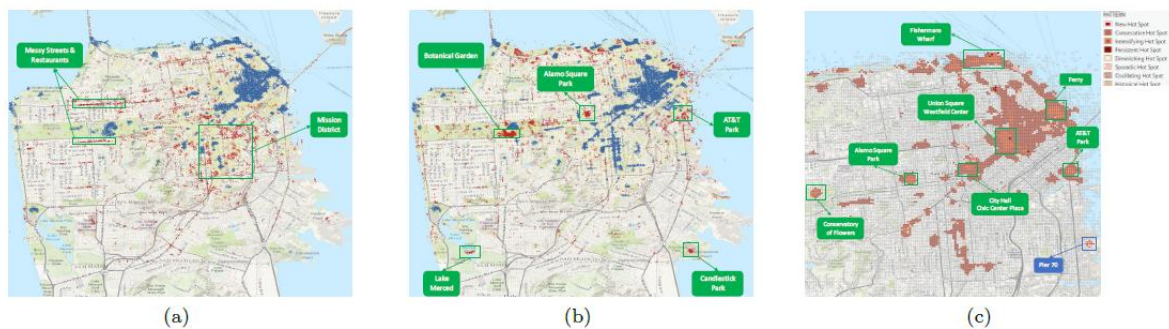


Figure 1: Spatial hotspot detection: (a) disgust; (b) joy. Red, yellow, blue represent hot, not significant and cold spots, respectively. Spatio-temporal hotspot detection: (c) joy. See the text for more details.

The figure included above is the only map in the article. A, B, and C have explanations in the subtext below the maps, which help the reader decipher the information. Clusters that the authors want the viewer to be aware of are boxed and labeled so the reader is easily aware of these important areas and can compare them to other areas. I think there are several areas for improvement. First, although the maps are described in the figure subtext, I think there isn't enough identifying information in each map to make the reader fully aware of what they're looking at. There is no north arrow, scale bar, or title for any of these maps. A and B appear to be at approximately the same scale, but C is definitely at a different scale – emphasizing the importance of a scale bar for orientation. Lastly, the map figures appear so small in the article that is hard to make out what the details; I am reading this article on the computer so I can zoom in very close, but if I were reading this on printed paper, I would have a hard time distinguishing any detail at all. In all, I think these maps would serve the article better if they were included separately, and with more detail, rather than compressed all together as they've done.